

PLamb Cosmology Model Summary Report

This report summarises the recent model fitting results using Pantheon+SH0ES supernovae (N=1701), BAO data, and variations of Λ CDM, KdS, and FR cosmologies.

The focus is on Bayesian Information Criterion (BIC) model comparison and variable interpretation.

1. Results Overview

Key findings from the latest runs:

- FR with supernova magnitude evolution ($\epsilon_M \approx -0.10$) significantly improves fit quality over flat Λ CDM ($\Delta\text{BIC} \approx -40$).
- Models with $\Lambda=0$ (pure KdS or FR no- Λ) are disfavoured due to BAO tension ($\Delta\text{BIC} > +150$).
- Curvature (Ω_k) alone does not provide a better fit than FR with ϵ_M .
- H_0 values cluster around 72–73 km/s/Mpc for FR/ Λ CDM, but drop to ~ 50 –53 for KdS solutions, which the data disfavour when BAO included.

2. Model-by-Model Breakdown

Flat Λ CDM (SN+BAO)

- H_0 : ≈ 73.39
- Ω_m : ≈ 0.263
- Ω_Λ : ≈ 0.737
- Ω_k : 0 (flat)
- ε_M : 0 (no evolution)
- BIC: ≈ 2113.55
- Comment: Baseline reference; decent fit but not optimal.

Curved Λ CDM (SN+BAO)

- H_0 : ≈ 73.19
- Ω_m : ≈ 0.224
- Ω_Λ : ≈ 0.667
- Ω_k : $\approx +0.11$
- ε_M : 0
- BIC: ≈ 2118.66
- Comment: Slight curvature improves χ^2 but BIC penalises.

FR varM log1pz (flat)

- H_0 : ≈ 72.95
- Ω_m : ≈ 0.252
- Ω_Λ : ≈ 0.748
- ε_M : ≈ -0.10
- BIC: ≈ 2070.73
- Comment: Strongest performer; $\Delta\text{BIC} \approx -43$ vs flat Λ CDM.

FR varM only (flat, with Λ)

- H_0 : ≈ 72.75
- Ω_m : ≈ 0.214
- Ω_Λ : ≈ 0.695
- ε_M : ≈ -0.10
- BIC: ≈ 2069.70
- Comment: Best balance; fits both SN and BAO.

FR varM only ($\Lambda=0$)

- H_0 : ≈ 70.1
- Ω_m : ≈ 0.0
- Ω_Λ : 0
- Ω_k : $\approx +1.0$
- ε_M : ≈ -0.087
- BIC: ≈ 2273.35
- Comment: Disfavoured; BAO χ^2 huge (≈ 230).

KdS flat (SN+BAO)

- H_0 : ≈ 52.75
- Ω_m : ≈ 0.40
- Ω_Λ : ≈ 0.60
- A_{acc} : ≈ 0.91
- n_{acc} : ≈ 0.82
- BIC: ≈ 2116.75
- Comment: High BAO χ^2 ; acceptance fraction low.

3. Variable Glossary

Below is a detailed glossary of all variables and parameters used:

- H_0 : Hubble constant today [km/s/Mpc], sets distance scale.
- Ω_m (Ω_m): Matter density fraction today, evolves as $(1+z)^3$.
- Ω_Λ (Ω_Λ): Cosmological constant density fraction.
- Ω_k (Ω_k): Curvature parameter, $\Omega_k = 1 - \Omega_m - \Omega_\Lambda$.
- χ^2 : Chi-square goodness of fit statistic.
- AIC: Akaike Information Criterion; penalises #params mildly.
- BIC: Bayesian Information Criterion; penalises #params strongly.
- A_{acc} : Amplitude of effective acceleration term in KdS/FR.
- n_{acc} : Index of acceleration term; implies effective w_{eff} .
- γ_c (γ_c): Variable speed of light slope: $c(z) = c_0(1 + \gamma_c z)$.
- ε_M (ε_M): Supernova absolute magnitude evolution parameter. $\Delta M(z) = \varepsilon_M f(z)$.
- μ (μ): Distance modulus; $\mu = 5 \log_{10}(D_L/10\text{pc}) + \Delta M(z)$.
- $E(z)$: Dimensionless Hubble parameter.
- D_L : Luminosity distance [Mpc].
- dM-mode: Choice of evolution form: linear_z or log1pz.